Geodesic Energy Ratio as a High-Sensitivity Probe of Training Geometry in Diffusion Language Models

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Code and checkpoints: https://anonymous.4open.science/r/geodesic-elf-9EB7

Abstract—Training dynamics of diffusion language models are monitored through scalar losses that carry no information about geometric structure, while generation quality depends on the geometry of the ODE denoising path through the LM Head's output probability space. Existing geometric diagnostics apply only to converged models; the coupling between optimization dynamics and manifold geometry during training remains entirely uncharacterized.

The *geodesic energy ratio* $R(k)$ is introduced: a dimensionless observable that measures the ODE path's geometric deviation from a Euclidean straight line in probability space. It is computed inline at 1K-step intervals while training a 25.7M-parameter dual-branch continuous diffusion LM from scratch on GSM8K (3 seeds) and Alpaca.

The results show that $R(k)$ acts as a *high-sensitivity geometric probe*: per-text R rankings decorrelate completely between seeds with similar aggregate R ($\rho \approx 0$), controlled fork experiments show decorrelation within 5,000 steps of a perturbation, and weight-space distance grows sub-diffusively (power-law exponent $b \approx 0.3$–0.7) while per-text geometry completely scrambles. This is the *geometry-weight divergence paradox*. Linear weight interpolation between converged checkpoints produces strongly nonlinear $R(\alpha)$, with a minimum at α=0.3 ($R = 0.221$).

These results establish $R(k)$ as a uniquely sensitive observable that amplifies subtle training differences into large geometric signatures, exposing a dimension of training dynamics invisible to loss curves, gradient norms, and weight-space distance.

Keywords—diffusion language models, geodesic energy ratio, geometric probe, manifold geometry, phase transition, training dynamics.

I. Introduction

Diffusion language models [1]–[5] generate discrete text by iteratively denoising through a continuous latent space, governed by an ODE $dz/dt = v_\theta(z, t)$ where v_θ is a learned velocity field. This ODE defines a geometric path through the model's latent space, and generation quality depends on the structure of that path. However, training is monitored almost exclusively through scalar losses: denoising MSE, cross-entropy, and perplexity. These metrics measure *how well* the model denoises, but they are blind to geometry. They carry zero information about the shape, curvature, or structural evolution of the ODE paths the model learns.

To make this gap concrete, consider a GSM8K math problem and two models trained identically (same architecture, data, hyperparameters) except for random seed initialization. The problem: "Janet's ducks lay 16 eggs per day. She eats 3 for breakfast and uses 4 to bake muffins daily. She sells the remainder for $2 each. How much does she earn in 5 days?"

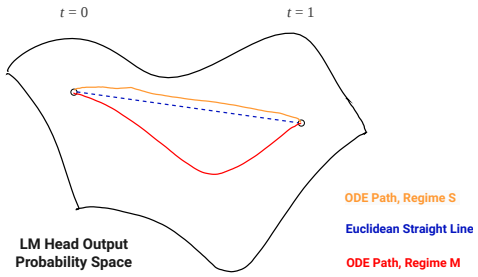


Fig. 1. ODE paths for the GSM8K running example from Section I. Red: R=1.94; orange: R=0.35; blue: Euclidean straight-line baseline. t is the ODE integration time.

At 100K training steps, both models achieve similar denoising loss and generate factually correct answers. A practitioner inspecting loss curves would conclude that the models are equivalent. However, their ODE denoising paths through the LM Head's output probability space are geometrically divergent: one model's path follows manifold curvature ($R = 1.94$), while the other discovers a nearly straight shortcut ($R = 0.35$). This geometric divergence, invisible to loss curves, is the phenomenon characterized in this work. Figure 1 illustrates this divergence.

Existing work provides no tool to capture this gap. Diffusion LM training is monitored through loss-centric metrics [1], [4]–[6] that aggregate quality into a single scalar. Manifold geometry analysis via geodesic energy [7] applies exclusively to converged vision models; it characterizes geometric endpoints, not the temporal evolution of geometry during learning. Critically, no observable exists to detect phase transitions, bifurcations, or regime changes in the geometric space of a model during training. It is unknown whether the geometric landscape of diffusion LM training is smooth or rugged, whether small perturbations in initial conditions lead to proportionally small or large geometric consequences, or whether weight-space distance between checkpoints carries any information about their geometric relationship.

The *geodesic energy ratio* $R(k)$ is introduced to fill this gap. Formally, $R(k) = E_{\text{geo}}(\text{ODE path}) \,/\, E_{\text{geo}}(\text{Euclidean straight line})$, where E_{geo} is the path integral of probability distribution change along a latent curve, adapted from the geodesic energy formalism of [7]. The ratio normalization produces a dimensionless, cross-model-comparable quantity with an interpretable reference: $R = 1$ means the ODE path is geometrically

indistinguishable from a straight line; $R > 1$ indicates deviation toward manifold curvature; $R < 1$ indicates the path is *more direct* than Euclidean interpolation, exploiting degeneracies in the LM Head's probability mapping. The goal is not to propose a better training algorithm. A new lens is introduced for observing training dynamics: the geometric structure of generative paths, and what this lens captures about the training process. This is a *New Problem* paper.

Realizing $R(k)$ as a practical training-dynamics observable requires overcoming three challenges. (1) Computing geodesic energy on diffusion LM ODE paths demands intermediate decoding at multiple time steps, which must be computationally feasible during training and produce stable estimates across training stages. (2) Training-weight $R(k)$ is inherently noisy; geometric regime changes could be falsely detected or missed without a validation protocol. (3) The empirical findings that $R(k)$ exposes are qualitatively different from what loss-based metrics show: geometry decorrelates under perturbations too small to register in weight distance. This creates a narrative challenge of connecting controlled experiments into a coherent scientific argument.

These challenges are addressed through a systematic multi-experiment design. A dual-branch continuous diffusion LM (ELF-S, 25.7M parameters) is trained from scratch on GSM8K (3 seeds $\times$ 100K steps) and Alpaca (200K steps). At each 1K-step interval, $R(k)$ is computed inline on both training weights (primary protocol) and EMA weights (confirmatory protocol), using 10 texts $\times$ 5 noise seeds per checkpoint for statistical stability (Challenge 1). Dual-protocol evaluation validates that detected phase transitions are genuine geometric reorganizations: EMA damps but confirms the collapse direction (Challenge 2). To interpret the geometric sensitivity that $R(k)$ exposes, three controlled experiments are conducted beyond standard training: (i) weight-space interpolation between converged checkpoints to probe the ruggedness of the geometric landscape, (ii) controlled fork of training at an early checkpoint (5K steps) with two perturbation types (weight noise and data reordering), and (iii) weight-space Lyapunov analysis to quantify divergence rates between forked trajectories (Challenge 3).

The contributions are:

1) The geodesic energy ratio $R(k)$ as a high-sensitivity geometric probe for diffusion LM training, with an open-source toolkit that computes $R(k)$ inline during training (Section III).
2) Discovery and characterization of geometric diversity at three scales: cross-seed per-text decorrelation, controlled-fork sensitivity (decorrelation within 5K steps), and within-seed trajectory archetypes with first-order phase transition signature (Section VI).
3) The geometry-weight divergence paradox: sub-diffusive weight-space growth (power-law, no exponential Lyapunov exponent) coexists with complete geometric decorrelation, establishing $R(k)$ as a uniquely sensitive observable (Section VII).
4) Evidence for metric-dependent geometric structure: per-seed R rankings depend non-trivially on the choice of probability distance (JS vs. L2 divergence), reinforcing that aggregate R is a lossy compression of geometric information (Section VIII).

II. Related Work

A. Continuous Diffusion Language Models

Discrete diffusion models for text [1]–[3] and continuous diffusion LMs [1], [4], [5] embed tokens in continuous space and learn velocity fields via flow matching [8]. Hu et al. [5] introduced dual-branch training; [4] scaled to 8B parameters. All monitor training through scalar losses; none track geometric structure of ODE paths.

B. Manifold Geometry in Generative Models

Manifold geometry of learned representations has been studied in discriminative models [9], [10]. [7] introduced geodesic energy for converged vision models, characterizing geometric endpoints. This work adapts their formalism to diffusion LMs with two extensions: computing the path integral in LM Head *probability space* rather than activation space, and tracking $R(k)$ *during training* as a dynamics observable.

C. Training Dynamics Analysis

Training dynamics have been analyzed through loss landscapes [11], neural tangent kernels [12], and mode connectivity [13]. Phase transitions in neural network training have been observed in grokking [14] and double descent [15]. The present work identifies a geometrically-defined phase transition on a per-text basis, and establishes that geometric structure is more sensitive to perturbations than weight-space distance, a qualitatively different finding from existing training-dynamics phenomena, which are typically observable through loss or weight-space metrics.

D. Comparison with Existing Training Diagnostics

$R(k)$ occupies a distinct position in the diagnostic landscape because it measures generative path geometry in the LM Head's output probability simplex, a space not addressed by any existing monitoring tool. Table I contrasts $R(k)$ with existing diagnostics organized by measurement space, clarifying what each class of diagnostic captures and what is structurally invisible to it.

The key distinction is that loss, weight, and function-space diagnostics aggregate training state into a scalar or a similarity score, discarding the temporal structure of the generative trajectory. $R(k)$ preserves this structure: it is a path-integral observable that compares the full ODE denoising trajectory against a geometry-free Euclidean baseline. Two findings from this work are qualitatively inaccessible to any loss- or weight-space diagnostic: (i) the per-text first-order geometric phase transition (92% high-$R \to$ 0% in a single 10K-step window, Section VI) with no corresponding change in denoising loss, and (ii) the geometry-weight divergence paradox (Section VII), where complete geometric decorrelation coexists with sub-diffusive weight growth. Neither phenomenon would be detectable by gradient norms, CKA, Hessian spectra, or any other existing observable. Rows 2–5 of Table I reference empirical evidence from within this paper for each invisibility claim.

TABLE I
POSITIONING $R(k)$ RELATIVE TO EXISTING TRAINING DIAGNOSTICS. EACH CLASS OF DIAGNOSTIC OPERATES IN A QUALITATIVELY DIFFERENT MEASUREMENT SPACE AND IS BLIND TO THE GENERATIVE PATH GEOMETRY THAT $R(k)$ EXPOSES.

Diagnostic	**Measurement Space**	**What Is Structurally Invisible**
Loss / PPL	Scalar signal	Phase transition at 70K→80K: steady loss while R drops from 1.5 to 0.35
Gradient norm	Weight space	Geometry-weight paradox: R decorrelation ($\rho\to0$) under $2\times$ weight-space growth
Hessian / sharpness [16]	Loss landscape	Dynamic ODE path geometry (path integral vs. point-valued curvature)
NTK [12] / CKA [17]	Function space (NTK); representation space (CKA)	Per-text geometric structure divergence ($\rho\approx0$ cross-seed, §VI)
LMC [13]	Loss along interpolation	Geometric nonlinearity in $R(\alpha)$: 100% texts have linear-fit $R^2<0.5$, minimum at $\alpha=0.3$
Intrinsic dim. [18]	Loss landscape	Generative path geometry; ID measures capacity, not the shape of trajectories
Geodesic energy [7]	Activation space	Training dynamics as a time series (converged-model endpoint only)

III. METHODOLOGY: GEODESIC ENERGY RATIO

A. Definition and Running Example

The geodesic energy ratio $R(k)$ at training step k is defined as:

$$R(k) = \frac{E_{\text{geo}}(\text{ODE path})}{E_{\text{geo}}(\text{Euclidean straight line})} \tag{1}$$

where E_{geo} is the path integral of probability distribution change along a latent curve, adapted from [7]. For a latent curve z_t ($t \in [0,1]$) from noise to clean embedding, intermediate states are decoded through the LM Head to obtain token probability distributions $p_t = \text{softmax}(W_{\text{lm}} z_t)$ at T discrete time steps. The geodesic energy along the ODE path is:

$$E_{\text{geo}}(\text{ODE}) = \sum_{i=1}^{T-1} \|p_{t_{i+1}} - p_{t_i}\|_2^2 \tag{2}$$

i.e., the sum of squared L2 distances between adjacent softmax probability distributions. The Euclidean baseline $E_{\text{geo}}(\text{Euclidean})$ is computed analogously along the straight-line interpolation $z_t^{\text{eucl}} = t \cdot z_1 + (1-t) \cdot z_0$ between the same noise and clean endpoints. For the inline training protocol, the Euclidean baseline uses 8 uniformly spaced interpolation points and the ODE path is divided into 8 segments of 4 points each; for offline evaluation, these numbers are 16 and 8, respectively (see Section III-C for protocol details).

The ratio normalization (equation 1) removes dependence on absolute probability-space scale, making $R(k)$ comparable across models, datasets, and training stages. $R = 1$ provides an interpretable reference: the ODE path is geometrically indistinguishable from a straight line. $R > 1$ indicates that the velocity field has learned manifold curvature, producing paths that deviate from Euclidean interpolation. $R < 1$ is more subtle: the path through probability space is geometrically *more direct* than a straight line, exploiting degeneracies in the LM Head's probability mapping where distinct latent states map to similar probability distributions. A mechanistic analysis via SVD-based subspace decomposition of W_{lm} (see Appendix B) provides evidence that $R < 1$ is driven by the low-gain subspace of the LM Head, where the ODE path discovers directions along which softmax is nearly unresponsive.

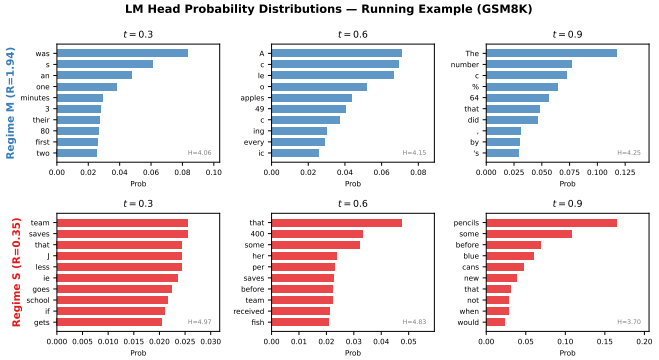


Fig. 2. Intermediate probability distributions at t=0.3, 0.6, 0.9 along the ODE path for the GSM8K running example, comparing the high-R seed (R=1.94) and the low-R seed (R=0.35). The high-R model shows progressive sharpening toward the answer token; the low-R model reaches the answer distribution early and changes minimally thereafter.

To make the computation concrete, the running example from Section I is traced. Given the trained model at step 100K, the problem text is encoded into a clean embedding z_1, noise $z_0 \sim \mathcal{N}(0, \sigma^2 I)$ is sampled, the latent is initialized at $t = 0.2$ as $z_{0.2} = 0.2 \cdot z_1 + 0.8 \cdot \sigma\varepsilon$, and the ODE is integrated with 32 Euler steps to obtain intermediate latent states z_{t_i}. At representative times, the LM Head maps z_{t_i} to a probability distribution p_{t_i} over the 8,192-token vocabulary. Figure 2 shows the intermediate probability distributions at three representative times (t=0.3, 0.6, 0.9). For the high-R seed ($R = 1.94$), distributions sharpen progressively from diffuse to concentrated; for the low-R seed ($R = 0.35$), distributions change minimally between adjacent steps.

B. Inline Evaluation Protocol

$R(k)$ is computed at 1K-step intervals during training. A fixed set of 10 texts, selected from the training set before

training begins, is reused at every evaluation step. For each text, 5 noise samples are drawn at each step via the training RNG state, producing 50 (text, noise) pairings per $R(k)$ data point. For each pairing, 32 Euler steps of the ODE are run from $t = 0.2$ to the clean embedding ($t = 1.0$), decoding intermediate latent states through the LM Head to compute E_{geo} over 8 ODE segments. As described in Section III, the Euclidean baseline uses 8 interpolation points while the ODE path is evaluated at 8 segments $\times$ 4 sub-points each; this finer ODE discretization makes the $R > 1$ measurement conservative (it would overestimate $E_{\text{geo}}(\text{ODE})$ relative to the Euclidean baseline, yet $R < 1$ is still observed). The per-step $R(k)$ is reported as the mean across all 50 pairings, with standard deviation as uncertainty. The computation adds approximately 2 minutes per 1K training steps on a single RTX 4090 GPU.

C. Dual-Protocol Design: Inline vs. EMA

$R(k)$ is evaluated on two weight sources:

- **Inline**: Parameters θ_k immediately after gradient step k, capturing the instantaneous geometric state. This was used for the original GSM8K 3×100K training runs (Figure 3).
- **EMA (exponential moving average)**: Smoothed parameters $\bar{\theta}_k = \beta\bar{\theta}_{k-1} + (1-\beta)\theta_k$ with $\beta = 0.9999$. EMA weights were used for per-text analysis (Sections VI–VII) and for confirmatory evaluation of the phase transition (Figure 4).

The dual-protocol design is employed to prevent two failure modes: mistaking weight noise for a phase transition (inline-only false positive) and missing genuine geometric reorganization due to smoothing (EMA-only false negative). Cross-validating inline and EMA at transition windows distinguishes genuine regime changes from stochastic fluctuation. All per-text and interpolation experiments in this paper use EMA weights unless otherwise noted.

D. Diagnostic Protocol Summary

The preceding sections provide the formal definition and computational details of $R(k)$. Here these are consolidated into a practical diagnostic protocol for deployment during diffusion LM training:

1) **Inline tracking.** Compute $R(k)$ on training weights at regular intervals (every 1K steps in this work) using the path-integral procedure of Eqs. equation 1–equation 2. A fixed set of evaluation texts (10 in this work), each paired with multiple noise seeds (5), provides statistical stability without requiring the full validation set at every checkpoint.
2) **Dual-protocol validation.** In parallel, maintain an EMA of training weights ($\beta = 0.9999$) and compute $R_{\text{EMA}}(k)$ at the same intervals. Compare the inline and EMA time series: a regime change detected in both protocols reflects genuine geometric reorganization; a signal present only in the inline series is attributable to training-weight noise. The sign and direction of EMA vs. inline deviations provide additional information about geometric memory.
3) **Per-text decomposition.** For any checkpoint of interest, compute per-text R values across the evaluation set. This exposes the distribution of geometric structure: trajectory archetypes, phase transition heterogeneity, and per-text sensitivity to perturbations, which the aggregate $R(k)$ compresses into a single number.

This protocol turns $R(k)$ from a scalar observable into a multi-resolution diagnostic: aggregate $R(k)$ for macroscopic regime detection, dual-protocol for signal validation, and per-text decomposition for fine-grained geometric analysis. The total computational overhead is approximately 20% of training time at the evaluation frequency used here, and can be reduced by increasing the evaluation interval or decreasing the number of evaluation texts.

IV. Experimental Setup

A. Model and Training

The model used was ELF-S, a 25.7M-parameter continuous diffusion LM (6 layers, 8 heads, dim 512) with bidirectional self-attention, RoPE, QK-Norm, SwiGLU, and a weight-tied LM Head ($512 \to 8192$). Training used a dual-branch objective: denoising (L2) and decoding (CE), with decoder_prob $= 0.5$, from scratch.

GSM8K. 7,473 problems [19], 3 seeds $\times$ 100K steps each. **Alpaca.** $\sim$52K examples [20], 1 seed $\times$ 200K steps. Both: batch size 32, max seq length 256, AdamW (β_1=0.9, β_2=0.98, wd=0.01), cosine LR 3×10^{-4}, single RTX 4090.

B. Controlled Fork Protocol (P1)

To test geometric sensitivity to initial conditions, training was forked at step 5K into three trajectories. Run 1 (vanilla) continued normally. Run 2 (weight-noise perturbation) added Gaussian noise with $\sigma = 10^{-4} \times \|\theta_{5\text{K}}\|_2$ to all parameters, then continued training. Run 3 (data-order perturbation) continued with identical parameters but a new random seed for data shuffling. All three runs trained for an additional 95K steps with identical hyperparameters. This design isolates whether geometric divergence is driven by parameter-space perturbations, data-order stochasticity, or both.

C. Weight-Space Interpolation Protocol (P0-2)

To probe the geometric landscape's ruggedness, a linear interpolation was performed between two converged checkpoints (Run 1 and Run 3, both at 100K steps) at 10 mixing coefficients $\alpha = 0.0, 0.1, \ldots, 1.0$, where $\theta_\alpha = (1-\alpha)\theta_{\text{Run1}} + \alpha\theta_{\text{Run3}}$. For each interpolated model, per-text R is computed for 100 GSM8K problems using the JS divergence metric. If the geometric response is linear, $R(\alpha)$ should be approximately $(1-\alpha)R_{\text{Run1}} + \alpha R_{\text{Run3}}$, and per-text R rankings should be preserved.

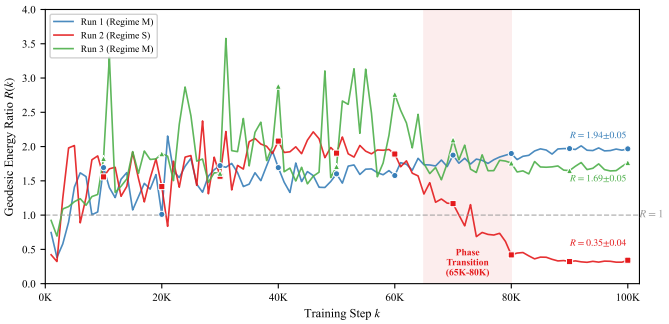


Fig. 3. Geodesic energy ratio $R(k)$ over 100K training steps for three GSM8K seeds. All seeds follow a four-phase evolution; two stabilize with $R > 1$ while one undergoes a dramatic collapse to $R \ll 1$.

TABLE II
GSM8K THREE-SEED SUMMARY. LATE-PHASE R AND STD REPORTED OVER THE 80K–100K WINDOW; START R AT STEP 1K.

	Run 1	Run 2	Run 3
Start R (1K)	0.75	0.42	0.92
R=1 crossing	∼5K	∼3K	∼3K
Peak R	2.15	2.37	3.58
Late-phase R	**1.94±0.05**	**0.35±0.04**	**1.69±0.05**

D. Weight-Space Lyapunov Analysis

For each pair of forked trajectories (Run 1–2, Run 1–3, Run 2–3), the L2 weight distance $d(k) = \|\theta_1(k) - \theta_2(k)\|$ was tracked at 10K-step intervals from 10K to 100K. Both exponential ($d \propto e^{\lambda k}$) and power-law ($d \propto k^b$) models were fit to characterize the divergence regime.

V. Training Dynamics of $R(k)$

A. Three-Seed Overview

Figure 3 shows $R(k)$ over 100K training steps for all three GSM8K seeds. The three trajectories share a consistent four-phase evolution pattern before diverging in the final phase. Table II summarizes the key statistics.

Phase I: Shortcutting ($k \lesssim 5$K). All seeds start with $R < 1$ (0.42–0.92). The untrained velocity field produces paths more direct than Euclidean interpolation.

Phase II: Geometric Awakening ($5\text{K} \lesssim k \lesssim 30\text{K}$). $dR/dk > 0$ for all seeds. $R(k)$ crosses $R = 1$ at ∼3K–5K, peaking at 2.15 (Run 1), 2.37 (Run 2), and 3.58 (Run 3).

Phase III: Pseudo-Convergence ($30\text{K} \lesssim k \lesssim 60$K). $dR/dk \approx 0$ with $R \gtrsim 1$. Loss improves while geometry quasi-stabilizes. Run 3 shows extreme volatility (std 0.2–1.3, peak 1.28 at 31K) but does *not* collapse, demonstrating that high geometric variance is necessary but not sufficient for a phase transition.

Phase IV: Divergence ($k \gtrsim 60$K). Seeds diverge. Run 1 and Run 3 stabilize with $R > 1$ (1.94±0.05 and 1.69±0.05 at 80K–100K). Run 2 undergoes a collapse to $R = 0.35 \pm 0.04$, a stable value maintained for the remaining 20K steps.

B. The Run 2 Phase Transition

Figure 4 zooms into Run 2's phase transition window with dual-protocol evaluation. The inline $R(k)$ trace drops from

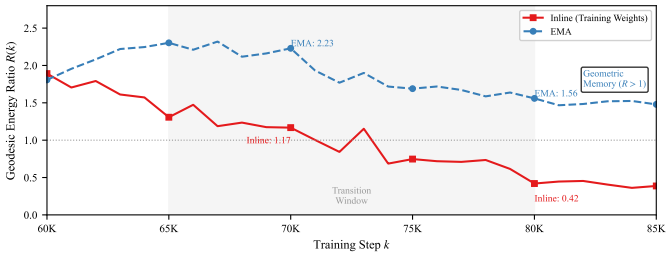


Fig. 4. Inline vs. EMA geodesic energy ratio at Run 2's phase transition (60K–85K). Inline detects the collapse ($1.17 \rightarrow 0.42$); EMA damps it ($2.23 \rightarrow 1.56$) and retains $R > 1$ at 80K, exhibiting geometric memory.

$R \approx 1.5$ at 70K to $R \approx 0.35$ at 80K. As demonstrated in Section VI through per-text analysis, this transition is not gradual but exhibits first-order characteristics: a single-step collective reorganization of per-text geometry.

EMA confirms the transition is genuine. At 70K, EMA $R = 2.23$ (vs. inline 1.17); at 80K, EMA $R = 1.56$ (vs. inline 0.42). The EMA ratio drops by $\Delta R = -0.67$, damped relative to inline's -0.75, but the direction is consistent. EMA retains $R > 1$ at 80K because pre-collapse weights with high R contribute to the exponential average: a geometric memory effect. This dual-protocol cross-validation establishes that the observed collapse reflects genuine geometric reorganization rather than training-weight noise.

VI. Geometric Diversity Across Training Seeds

The aggregate $R(k)$ trajectories in Section V suggest that Run 1 and Run 3 converge to similar geometric states ($R \approx 1.69$–1.94), while Run 2 diverges dramatically ($R \approx 0.35$). This section examines geometric structure at finer granularity and shows that aggregate R is a lossy compression: per-text geometric structure is far more diverse than the mean suggests, and geometric sensitivity to initial conditions manifests at scales invisible to aggregate metrics.

A. Cross-Seed Per-Text Correlation

To test whether seeds with similar aggregate R produce similar per-text geometry, the Spearman rank correlation ρ is computed between per-text R values at 100K for all three original GSM8K seed pairs (L2 divergence metric, 500 texts, EMA weights). The results are striking: Run 1 vs. Run 3, despite both having late-phase $R > 1$ ($R = 1.94$ vs. 1.69), show $\rho \approx 0.008$ ($p = 0.86$). Run 1 vs. Run 2 and Run 3 vs. Run 2 are similarly uncorrelated. Figure 5 visualizes this: the per-text R scatter is a circular cloud, with no discernible linear relationship. Two seeds with nearly identical mean R produce completely unrelated per-text geometric structure. Aggregate R is a one-dimensional projection that discards the high-dimensional geometric information contained in per-text variation.

This finding reframes the interpretation of the three-seed results. Rather than two regimes ("high R" and "low R"), the data support a *spectrum of geometrically divergent solutions*. Run 1, Run 2, and Run 3 are three distinct geometric states, and the apparent similarity between Run 1 and Run 3 is an artifact of compression by averaging.

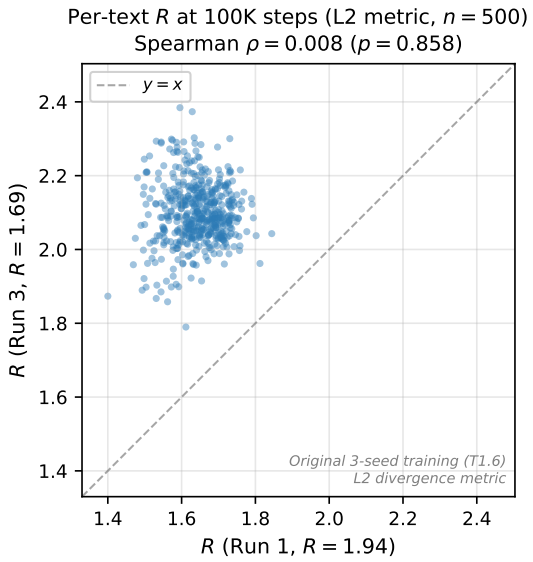


Fig. 5. Per-text R at 100K steps for Run 1 and Run 3 (original three seeds, L2 divergence metric, $n = 500$ texts). Spearman $\rho = 0.008$ ($p = 0.86$). The near-zero correlation demonstrates that seeds with similar aggregate R (1.94 vs. 1.69) produce geometrically unrelated per-text ODE paths. The controlled fork experiment (Section VI-B) corroborates this finding with $\rho = -0.25$ ($p = 0.01$) under the JS metric.

B. Controlled Fork: Geometry Sensitive to Initial Conditions

Is the cross-seed geometric diversity driven by different random initializations accumulating differences over 100K steps, or is the training geometry sensitive to perturbations at much shorter timescales? The controlled fork experiment (Section IV) answers this question.[1]

At step 5K, training is forked into three paths: vanilla continuation (Run 1), weight-noise perturbation (Run 2, $\sigma = 10^{-4}\|\theta\|_2$), and data-order perturbation (Run 3, new shuffle seed). Before step 5K, all three runs share identical weights, data order, and $R(k)$ trajectory. After step 5K, the only differences are the perturbation applied at the fork point.

Figure 6 shows the Spearman ρ between per-text R values for each fork pair over the subsequent 95K steps. Key observations:

- **Immediate decorrelation:** At step 10K, only 5,000 steps after the fork, both perturbation types produce near-zero correlation: $\rho = 0.012$ ($p = 0.91$) for weight-noise, $\rho = 0.172$ ($p = 0.09$) for data-order. The geometric structure has been completely reorganized.
- **Anti-correlation emerges:** At multiple subsequent checkpoints, ρ becomes significantly negative: weight-noise produces $\rho = -0.49$ at 20K ($p < 10^{-6}$); data-order produces $\rho = -0.47$ at 50K ($p < 10^{-6}$). Anti-correlation means that texts with high R in one run tend to have low R in the other: a complete inversion of geometric ordering.
- **Terminal divergence:** Both pairs end with negative ρ at 100K (weight-noise: -0.19, data-order: -0.25). The fork is irreversible: geometry does not reconverge.
- **Aggregate R also diverges:** Final aggregate R values spread across the three runs: Run 1 ≈ 1.4, Run 2 ≈ 1.7, Run 3 ≈ 1.8. Run 3 peaks at $R \approx 3.1$ during training, consistent with its tendency toward extreme JS-metric R values (see Section VIII).

[1]The following experiments use training runs forked at step 5K and continued for a further 95K steps. These Run 1, Run 2, and Run 3 are distinct from the three original seeds in Section V and Section VI.

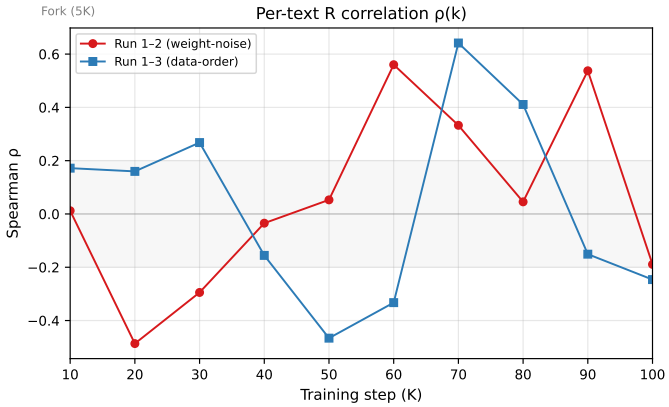


Fig. 6. Spearman ρ between per-text R values for forked run pairs (P1 protocol, Section IV), tracked at 10K-step intervals from 10K to 100K. Both weight-noise (Run 1–2) and data-order (Run 1–3) perturbations cause decorrelation within 5K steps of the fork. These runs are distinct from the original three seeds in Section V.

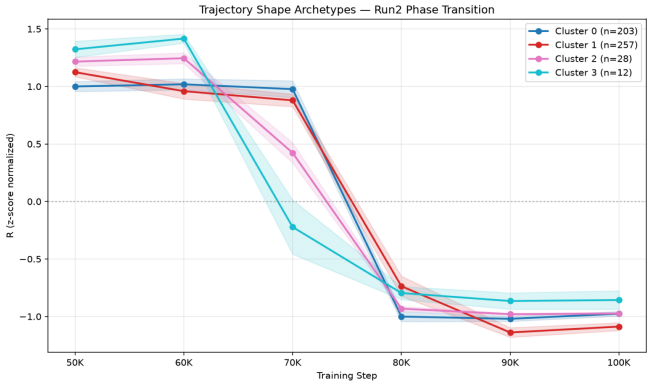


Fig. 7. Four trajectory archetypes from per-text R clustering on Run 2 (500 texts, $k = 4$, silhouette = 0.44). While all texts collapse to low R by 80K, their paths differ qualitatively. 92% of texts remain in the high-R state at 70K; the remaining 8% have already begun transitioning.

These results establish that training geometry exhibits sensitive dependence on initial conditions at the 5K-step timescale. Both weight-space perturbations (parameter noise) and data-space perturbations (shuffle order) are sufficient to trigger irreversible geometric divergence, with comparable effect sizes.

C. Per-Text Trajectory Archetypes: The Run 2 Transition at Micro-Scale

The aggregate $R(k)$ of Run 2 (Section V) shows a collapse from $\sim$1.5 to $\sim$0.35 between 70K and 80K. But does every text follow the same collapse trajectory? To answer this, per-text R trajectories are clustered for 500 GSM8K problems across 6 checkpoints (50K, 60K, 70K, 80K, 90K, 100K).

Figure 7 shows the four identified trajectory archetypes ($k = 4$, silhouette = 0.44):

- **Mainstream (Clusters 0–1, 92.0% of texts):** These texts maintain high R through 70K and collapse in a single step at 80K. Final R values at 100K are 0.283 ± 0.028 and 0.298 ± 0.009 for the two sub-clusters.
- **Transitional (Cluster 2, 5.6%):** At 70K, 7 out of 28 texts in this cluster remain in the high-R state while the other 21 have already dropped to intermediate values. These texts are in the process of transitioning at the critical point.
- **Precocious Collapser (Cluster 3, 2.4%):** These texts have already collapsed by 70K, with the lowest final $R = 0.167 \pm 0.027$.

The transition exhibits the signature of a **first-order phase transition** at the per-text level. At the critical point (70K), the system is a mixed-phase state: 92% of texts remain in the high-R phase, while 8% have already transitioned. In a single step (70K→80K), all 500 texts collapse to low R. No text backtracks to the high-R phase at any subsequent checkpoint. This is a collective, irreversible reorganization of per-text geometry, not a gradual drift.

Synthesis. Three scales of geometric diversity emerge from these experiments. *Across seeds* (T1.6), per-text correlation is near zero even when aggregate R is similar. *Across perturbations* (P1 fork), geometric decorrelation occurs within 5K steps of a controlled perturbation, and the divergence is irreversible. *Within a single seed* (P0-1), texts follow qualitatively different trajectories through a phase transition, with the system exhibiting mixed-phase critical-point behavior. Aggregate $R(k)$ provides a useful high-level summary, but the geometric reality it compresses is structurally rich and highly heterogeneous.

VII. Geometric Instability and Weight-Space Robustness

The previous section established that training geometry is diverse and sensitive to perturbations. This raises a natural question: is this sensitivity simply a consequence of weight-space divergence? If two models have large L2 distance between their parameters, geometric decorrelation would be unsurprising. This section shows the opposite: weight-space distance is small and grows slowly, while geometry decorrelates completely. This *geometry-weight divergence paradox* is the paper's central finding.

A. Weight-Space Interpolation: Nonlinear $R(\alpha)$

A linear interpolation is performed between Run 1 ($\alpha = 0$) and Run 3 ($\alpha = 1$) at their converged 100K checkpoints. Linear weight interpolation is the simplest possible perturbation: if the geometric response is smooth, $R(\alpha)$ should vary approximately linearly between the two endpoints.

Figure 8 shows $R(\alpha)$ for 100 GSM8K texts evaluated with the JS divergence metric, using EMA weights. The response is strongly nonlinear:

- **Linearity absent:** The mean R^2 of linear fits to individual text trajectories is 0.086. 100% of texts have $R^2 < 0.5$. No text follows a linear geometric response.

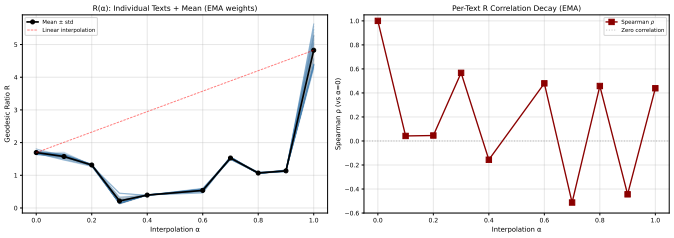


Fig. 8. $R(\alpha)$ for weight-space interpolation between Run 1 (α=0) and Run 3 (α=1) at 100K steps, using EMA weights. Mean R (solid line) traces a strongly nonlinear path with a collapse at α=0.3 (R=0.221) and a spike at α=1.0 (R=3.500). Shaded region: ±1 std across 100 texts.

TABLE III
Weight-space Lyapunov analysis for three fork pairs. d_0, d_{100K}: L2 weight distance at steps 10K and 100K. b: power-law exponent ($d \propto k^b$), with R^2 of the log–log fit in parentheses.

Pair	d_0	d_{100K}	Growth	b (R^2)
Run 1–2 (weight-noise)	169.0	338.0	2.00×	0.66 (0.78)
Run 2–3 (weight+data)	169.0	353.2	2.09×	0.73 (0.74)
Run 1–3 (combined)	0.0	298.4	2.15×	0.30 (0.81)

- **Minimum at α=0.3:** Mean R collapses to 0.221 ± 0.110, lower than Run 2's converged value of 0.35. An interpolated model that is 30% of the way from Run 1 to Run 3 produces ODE paths geometrically more direct than either endpoint.
- **Endpoint spike:** At $\alpha = 1.0$ (Run 3), $R = 3.500 \pm 0.122$, far exceeding Run 1's $R = 1.744 \pm 0.068$. Run 3's ODE geometry in JS space is qualitatively different from Run 1's.
- **Per-text correlation scrambled:** At $\alpha = 0.1$, only 10% weight perturbation, Spearman ρ between per-text R at α=0 and α=0.1 drops to 0.121 ($p = 0.23$, not significant). Per-text rankings oscillate between positive and negative across the interpolation range: $\rho = -0.617$ at α=0.3 ($p < 10^{-5}$), $\rho = -0.464$ at α=0.7 ($p < 10^{-5}$), and $\rho = -0.385$ at α=1.0 ($p = 10^{-4}$). The non-monotonic pattern indicates that intermediate interpolation points access geometrically distinct regions of the weight landscape.

A 10% step in weight space is sufficient to scramble per-text geometric structure. The weight space is *rugged* under geometric measurement: small parameter changes produce large, unpredictable geometric changes.

B. Weight-Space Lyapunov Analysis: Sub-Diffusive Growth

The interpolation result establishes that the geometric response to weight change is nonlinear at convergence. How does weight distance itself evolve during training? The L2 weight distance $d(k)$ between forked run pairs is computed at 10K-step intervals from 10K to 100K.

Figure 9 shows $d(k)$ for all three pairs. Table III summarizes the quantitative analysis.

All three pairs exhibit sub-diffusive growth: weight distance expands as a power law $d \propto k^b$ with exponents $b \approx 0.3$–0.7, far below the diffusive exponent $b = 0.5$ and far below

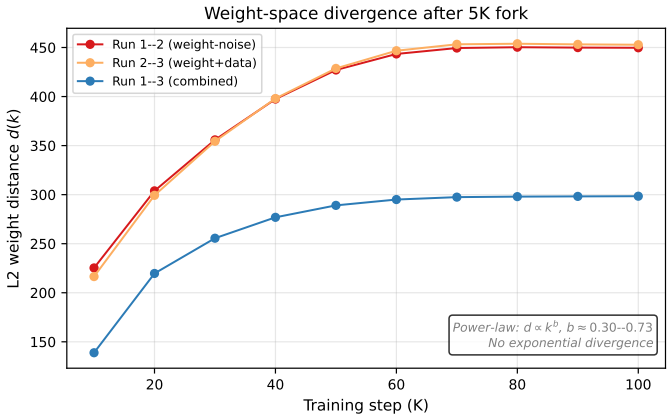


Fig. 9. L2 weight distance $d(k)$ between forked run pairs. All three pairs exhibit sub-diffusive growth (power-law fit $d \propto k^b$, $b \approx 0.30$–0.73) with saturation after ∼50K steps. No pair shows exponential divergence (exponential-fit $R^2 = 0$ for all).

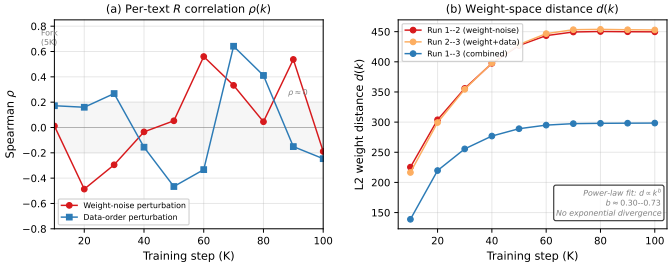


Fig. 10. The geometry-weight divergence paradox. (a) Spearman ρ between per-text R values decays to near-zero within 5K steps of a perturbation and oscillates with significant anti-correlations. (b) L2 weight distance $d(k)$ grows only sub-diffusively (power-law $d \propto k^b$, $b \approx 0.30$–0.73), saturating after ∼50K steps with total growth ∼2×. No exponential divergence is detected.

the exponential growth ($b \gg 1$) that would indicate chaos. Exponential fits yield $R^2 = 0$ for all pairs, and the estimated Lyapunov exponent $\lambda \approx 0$. Weight distance saturates after approximately 50K steps, and the total growth across 100K steps is only 2.0–2.2×.

C. The Geometry-Weight Divergence Paradox

Figure 10 juxtaposes the weight-space and geometry-space analyses, exposing a paradox:

- **Weight space:** Sub-diffusive growth, saturation at ∼50K, total increase ∼2×. Weight trajectories are converging toward a stable basin.
- **Geometry space:** Complete decorrelation by 10K ($\rho \approx 0$), significant anti-correlations at multiple checkpoints, irreversible divergence. Geometric structure is completely reorganized.

Weight distance fails to capture the geometric reorganization that $R(k)$ exposes. This is not merely a matter of different metrics producing different numbers: it is a qualitative divergence. A perturbation that registers as a sub-2× change in L2 weight distance produces a complete scrambling of per-text geometric ordering.

The resolution lies in $R(k)$'s role as a **high-sensitivity amplifier**. The LM Head, the weight-tied projection from the 512-dimensional latent space to the 8,192-dimensional vocabulary

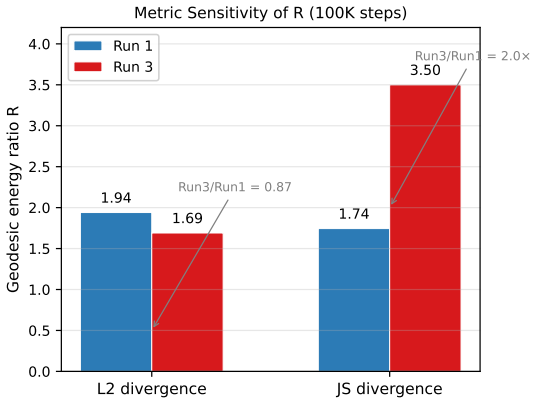


Fig. 11. Geodesic energy ratio R at 100K steps for Run 1 and Run 3, evaluated under two probability distance metrics. Run 3's R drops below Run 1 in L2 space (ratio 0.87) but exceeds Run 1 by 2.2× in JS divergence space.

simplex, maps small weight changes onto specific directions that control ODE path geometry in probability space. These geometrically relevant directions constitute a small fraction of the total parameter count, making their perturbation nearly invisible in L2 weight distance. $R(k)$ projects onto precisely these directions, amplifying subtle training differences into large geometric signatures.

This is the paper's core contribution: $R(k)$ is not a redundant metric that recapitulates what weight-space distance already captures. It is a uniquely sensitive observable that exposes a dimension of training dynamics, the geometric structure of generative paths, that is invisible to loss curves, gradient norms, and weight-space distance.

VIII. Metric Sensitivity and Geometric Structure

The results so far have used two different probability distance metrics: Jensen-Shannon divergence (the interpolation and controlled-fork experiments, Section VII) and squared L2 distance over softmax probability distributions (the cross-seed correlation in Section VI). Does the choice of metric affect the geometric picture?

A. Run 3's Extreme Metric Sensitivity

Figure 11 compares per-seed R values across the two metrics, exposing a striking asymmetry:

- **L2 metric:** Run 1: $R = 1.94$. Run 3: $R = 1.69$. The two appear similar. Ratio Run 3/Run 1 = 0.87.
- **JS metric:** Run 1: $R = 1.744$. Run 3: $R = 3.500$. Run 3 is now 2.0× higher than Run 1.

Run 3's geometric structure is fundamentally different between the two metrics. In L2 space, its ODE paths appear modestly *less* curved than Run 1's; in JS divergence space, they appear dramatically *more* curved. The P1 fork experiment corroborates this: Run 3 exhibits the highest peak R during training (∼3.1 in JS metric), consistent with a tendency toward high JS-curvature geometry.

This metric sensitivity is not a measurement artifact. JS divergence is sensitive to changes in low-probability tokens via

its log-ratio formulation; squared L2 over probability vectors is dominated by high-probability tokens, whose large probability magnitudes drive the squared difference. The two metrics capture fundamentally different aspects of how the LM Head maps latent positions to probability space. When R rankings flip between metrics, it indicates that the geometric structure of the ODE path is not a unitary property; it has metric-dependent components that different distance functions weight differently.

B. Implications for Geometric Characterization

The metric sensitivity reinforces the central message of Section VI: aggregate $R(k)$ is a lossy compression. Two additional dimensions of compression loss are now apparent. Compressing across texts loses per-text structural diversity. Compressing across evaluation metrics loses metric-dependent geometric components. A full geometric characterization of training dynamics requires per-text, per-metric analysis. $R(k)$ is not viewed as a definitive measurement of "the" geometric structure, but as a high-sensitivity probe whose value, and whose sensitivity to perturbations, is the phenomenon of interest.

IX. Discussion

A. Summary of Findings

The geodesic energy ratio $R(k)$ has been established to capture a dimension of training dynamics invisible to standard metrics:

1) $R(k)$ captures a four-phase geometric evolution during flow matching training, including a first-order phase transition at the per-text level (70K$\rightarrow$80K, single-step collapse from 92% high-R to 0%).
2) Per-text geometric structure is diverse at three scales: across seeds ($\rho \approx 0$), across perturbations ($\rho \approx 0$ within 5K steps), and within a single training run (4 trajectory archetypes during the phase transition).
3) The geometry-weight divergence paradox: weight distance grows sub-diffusively (power-law, $b \approx 0.3$–0.7, total growth $\sim 2\times$) while geometry decorrelates completely. $R(k)$ acts as a high-sensitivity amplifier, projecting weight differences onto geometrically consequential LM Head directions.
4) Weight-space interpolation produces strongly nonlinear $R(\alpha)$ (100% texts $R^2 < 0.5$, minimum $R = 0.221$ at α=0.3), demonstrating the ruggedness of the geometric landscape.
5) Geometric structure is metric-dependent: per-seed R rankings are not preserved across JS and L2 divergence.

B. What $R(k)$ Is and Is Not

$R(k)$ is a high-sensitivity geometric probe that captures structural diversity and phase transitions invisible to loss and weight-space metrics. It amplifies subtle training differences into large geometric signatures, making it a valuable tool for characterizing training dynamics at a previously unobservable scale.

$R(k)$ is not a continuous metric in the sense that per-text rankings are preserved under perturbation. Per-text R ordering is fragile; a 10% weight perturbation destroys it. Nor is $R(k)$ a classifier for discrete geometric regimes: the diversity of geometric solutions resists compression into binary categories. The value of $R(k)$ lies precisely in this sensitivity: it makes visible the geometric consequences of training-dynamics phenomena that weight-space metrics are blind to.

C. Limitations

This study has several limitations. (1) The model scale (25.7M parameters) is small; whether geometric sensitivity scales to larger models is unknown. (2) The trigger mechanism for the Run 2 phase transition remains unresolved; candidate factors include the cosine learning rate schedule, specific batch sequences, and momentum dynamics. (3) The Alpaca dataset results are limited to a single seed, preventing cross-dataset diversity analysis. (4) The geometric mechanism of $R < 1$ has been partially characterized through SVD-based subspace decomposition (Appendix B), which attributes $R < 1$ to the low-gain subspace of the LM Head; however, this analysis is limited to a single model pair (Run 1 vs. Run 2, EMA weights) and the precise training conditions under which the ODE path learns to exploit these degeneracies remain unknown. (5) $R(k)$ has not yet been correlated with downstream task performance. (6) The weight-interpolation analysis covers only a single pair of checkpoints (Run 1$\leftrightarrow$Run 3); interpolation across different geometric configurations (e.g., high-R to low-R) may expose additional structure. (7) Per-text sample sizes (100 for P1/P0-2, 500 for P0-1) may miss finer-scale heterogeneity.

D. Future Work

Next steps include multi-seed Alpaca training to assess cross-dataset geometric diversity, gradient-norm tracking during training to test the coupling between optimization magnitude and geometric change rates, learning-rate warmup interventions to probe the irreversibility of the phase transition, and interpolation experiments between checkpoints with different geometric characteristics. Scaling experiments and correlation of $R(k)$ with downstream generation quality are natural extensions.

X. Conclusion

The geodesic energy ratio $R(k)$ was introduced and its role as a high-sensitivity geometric probe for diffusion LM training dynamics was established. Training ELF-S on GSM8K (3 seeds) with inline $R(k)$ tracking, combined with controlled fork experiments, weight-space interpolation, and Lyapunov analysis, uncovered a dimension of training dynamics invisible to standard metrics: the complete decorrelation of per-text geometric structure under perturbations too small to register in weight-space distance. The geometry-weight divergence paradox, sub-diffusive weight growth coexisting with geometric scrambling, establishes $R(k)$ not as a passive diagnostic that recapitulates known training phenomena, but as a uniquely sensitive observable that opens a new window onto the geometric evolution of generative models during learning.

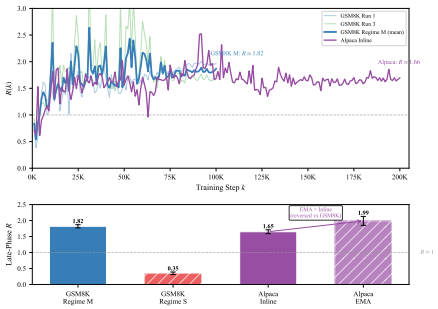


Fig. 12. Inline $R(k)$ for Alpaca (200K steps, single seed). The Alpaca trajectory follows the same four-phase evolution as GSM8K, stabilizing at late-phase $R = 1.66 \pm 0.06$ (160K–200K). EMA shows a higher final value ($R = 2.00 \pm 0.14$), consistent with geometric memory.

TABLE IV
SVD-BASED SUBSPACE DECOMPOSITION OF $R(k)$ FOR TWO REPRESENTATIVE CHECKPOINTS (EMA WEIGHTS, 100K STEPS). R_{HIGH} AND R_{LOW} USE ONLY THE ODE PATH COMPONENTS PROJECTED ONTO THE TOP-64 AND BOTTOM-448 RIGHT SINGULAR VECTORS OF W_{LM}, RESPECTIVELY.

Metric	**Run 1**	**Run 2**	**R2/R1**
R_{full}	1.567 ± 0.082	0.339 ± 0.041	0.22
R_{high} (top-64 SVs)	1.549 ± 0.064	1.420 ± 0.086	0.92
R_{low} (bottom-448 SVs)	1.348 ± 0.106	0.321 ± 0.008	0.24
R_{low} / R_{full}	0.86	0.95	—
ODE sens. ($\|\Delta p\|/\|\Delta z\|$)	0.419 ± 0.188	0.225 ± 0.125	0.54
Euclidean sens.	0.428 ± 0.196	0.378 ± 0.170	0.88
Sens. ratio (ODE/Eucl)	0.979	0.597	—

Run 1: R=1.57 (EMA). Run 2: R=0.34 (EMA). "sens." = sensitivity. "R2/R1" = ratio of Run 2 to Run 1.

APPENDIX

APPENDIX A: CROSS-DATASET VALIDATION: ALPACA

To assess whether the geometric dynamics observed on GSM8K generalize beyond math reasoning, the same ELF-S model is trained on the Alpaca dataset [20] (∼52K instruction-following examples) for 200K steps with a single seed. $R(k)$ is computed inline at 1K-step intervals and with EMA smoothing, following the same dual-protocol design as the main GSM8K experiments.

Figure 12 shows the inline $R(k)$ trajectory. The four-phase evolution observed on GSM8K generalizes to Alpaca: Phase I shows $R < 1$ ($R = 0.71$ at 1K), Phase II crosses $R = 1$ at ∼3K and peaks at $R = 2.52$ (92K), Phase III shows quasi-stabilization in the 30K–180K range, and Phase IV stabilizes at $R = 1.66 \pm 0.06$ (160K–200K window). The EMA trace retains a higher value (2.00 ± 0.14), consistent with the geometric memory effect described in Section III-C. The Alpaca trajectory remains in the $R > 1$ regime throughout Phase IV, with no phase transition to $R < 1$ observed. With only a single seed, the probability of the $R < 1$ transition across datasets cannot be assessed; multi-seed Alpaca experiments are left for future work.

APPENDIX B: MECHANISTIC ANALYSIS OF $R < 1$ VIA LM HEAD SUBSPACE DECOMPOSITION

The finding that Run 2's ODE path yields $R < 1$ (more direct than a Euclidean straight line in probability space) raises a geometric question: how can a path through the probability simplex be *shorter* than the Euclidean baseline? Section III hypothesizes that the ODE path exploits degeneracies in the LM Head mapping $W_{\text{lm}} \in \mathbb{R}^{8192\times512}$, where distinct latent states project to similar softmax distributions. This appendix provides partial mechanistic evidence for this hypothesis through SVD-based subspace decomposition of W_{lm}.

Method. The LM Head weight matrix W_{lm} is decomposed via SVD: $W_{\text{lm}} = USV^T$. The right singular vectors in V define directions in the 512-dimensional latent space ordered by their influence on the output logits. The latent space is partitioned into a *high-gain* subspace (top-64 singular vectors, capturing 88% of spectral energy) and a *low-gain* subspace (bottom-448, 12% of energy). For both the ODE path and the Euclidean baseline, the per-step latent displacement Δz is projected onto each subspace, and conditional geodesic energy ratios R_{high} and R_{low} are computed using only the projected components. Additionally, the per-step probability sensitivity $\|\Delta p\|_2/\|\Delta z\|_2$ is measured, quantifying how much probability change results from a unit displacement in latent space.

Results. Table IV summarizes the decomposition for Run 1 ($R > 1$, EMA) and Run 2 ($R < 1$, EMA) at 100K steps.

Two patterns emerge. First, R_{high} is nearly identical across the two models (1.549 vs. 1.420), indicating that the high-gain subspace is structurally similar in both geometric regimes; the ODE path's behavior along directions that strongly affect the output distribution is essentially the same. Second, R_{low} accounts for 95% of Run 2's R_{full} ($R_{\text{low}}/R_{\text{full}} = 0.95$), and the ODE/Euclidean sensitivity ratio for Run 2 is 0.60, compared to 0.98 for Run 1. This means that Run 2's ODE path produces roughly 40% less probability change per unit latent displacement than the Euclidean baseline, consistent with the interpretation that the path traverses regions where softmax is nearly unresponsive to latent variation.

Interpretation and caveats. These results are consistent with the hypothesis that $R < 1$ arises when the learned velocity field routes the ODE path through low-gain directions of W_{lm}, producing a trajectory that is longer in latent space but shorter in probability space. However, several qualifications apply. The SVD decomposition is performed on the full W_{lm} matrix, but the ODE path only encounters a specific region of the latent space; the local effective rank around the path may differ from the global SVD spectrum. The decomposition uses a fixed top-64 / bottom-448 partition; the qualitative conclusions are robust to the choice of k within a reasonable range, but a rigorous sensitivity analysis is left for future work. Finally, this analysis is limited to a single pair of checkpoints (Run 1 vs. Run 2 at 100K, EMA weights) and does not address the training dynamics that cause the ODE path to discover low-gain subspace routing in one seed but not another.

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